



WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**
WPS No. **WPS-RNC-PN1-S** Date **6-Jul-13** Revision No. **0**
Supporting PQR No.(s) **KWC-06A0017-PQR-RNC-PN1** Date **23-Feb-13** Type(s) **Manual**
Welding Process(es) **Shield Metal Arc welding (SMAW)** (Automatic, **Manual**, Machine, or Semi-Auto.)

JOINT (QW-402)

Joint Design **Groove/Fillet**

Backing (Yes) **(No)** **X**

Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes _____

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **1** Group No. **1 or 2** to P-No. **1** Group No. **1 or 2**

Material Specification Type and Grade **A106 Gr.B, A105 Gr.B, A 234 WPB (or as per P-1)** TO **A106 Gr.B, A105 Gr.B, A 234 WPB (or as per P-1)**

Chem. Analysis and Mech. Prop. _____

Thickness Range :

Base Metal : Groove **5 mm -15.94 mm.** Fillet **All**

Pipe Dia. Range : Groove **NPS 2⁷/8"-Unlimited** Fillet **All**

Other **Maximum Pass Thickness 1/2 in. (13 mm) for SMAW**

FILLER METALS (QW-404)

SMAW

Spec. No. (SFA) **5.1**

AWS No. (Class) **E 7016**

F-No. **4**

A-No. **1**

Size of Filler Metals **Ø 3.2 & 4.0 mm.**

Deposited Weld Metals **7.97 mm.**

Thickness Range :

Groove **Max. 15.94 mm.**

Fillet **ALL**

Electrode-Flux (Class) **-**

Flux Trade Name **-**

Consumable Insert **-**

Other **Kobelco LB 52 or Equivalents**

POSITION (QW-405)

Position(s) of Groove **ALL**

Welding Progression Up **X** Down _____

Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **N/A**

Time Range **N/A**

Notes **N/A**

PREHEAT (QW-406)

Preheat Temp. Min. **30°C (Ambient)**

Interpass Temp. Max. **250°C**

Preheat Maintenance **N/A**

Notes _____

GAS (QW-408)

Percent Composition

	Gas(es)	(Mixture)	Flow Rate
Shielding	-	-	-
Trailing	-	-	-
Backing	-	-	-

WPS No. WPS-RNC-PN1-S Rev. 0

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☒ AC ☒ DC Polarity SEE BELOW
 Amps (Range) SEE BELOW Volts (Range) SEE BELOW
 Tungsten Electrode Size and Type N/A
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS
 Orifice or Gas Cup Size N/A
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
 Method of Back Gouging AIR ARC GOUGING OR GRINDING (IF REQUIRED)
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE
 Travel Speed (Range) SEE BELOW
 Peening N/A
 Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	SMAW	E 7016	3.2 & 4.0	DC EP/AC	90-120	20-38	5-15.24	
2nd & Others	SMAW	E 7016	3.2 & 4.0	DC EP/AC	90-120	20-38	5-15.24	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of ASME IX, 2010 latest Revision.

Prepared by :

Mr. Kittipong M.
Name

Signed

6/30/13
Date

Reviewed by :

Phuchin S.
Name

Signed

6/7/13
Date

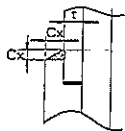
Approved by :

Phuchin S.
Name

Signed

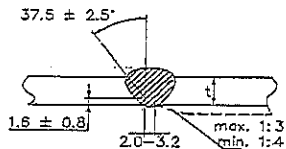
6/7/13.
Date

ATTACHED FIGURE JOINT SKETCH



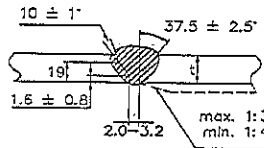
Cx (min.) = $1.25t$ but not less than 3.2 mm.

JOINT No. A1



Misalignment (max.) = 1.5 mm.

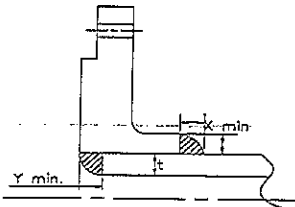
JOINT No. B1 & B2



Approx. 1.6 mm.

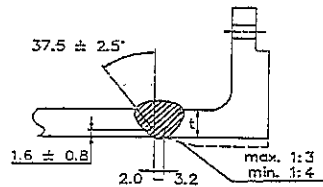
X min. = The lesser of $1.4t$ or thickness of the hub.

JOINT No. A2



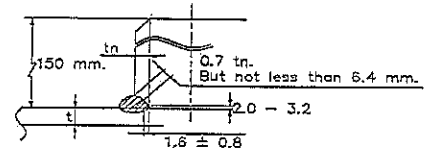
X min. = The lesser of $1.4t$ or thickness of the hub.
Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



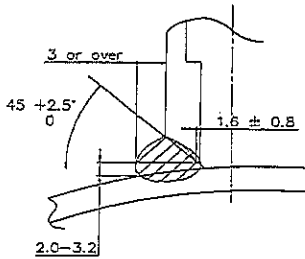
Misalignment (max.) = 1.5 mm.

JOINT No. B3

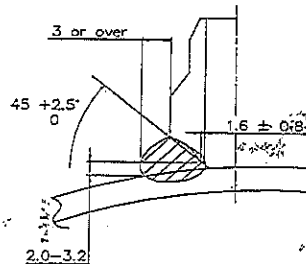


Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

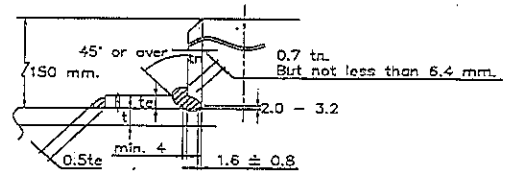
JOINT No. B4



JOINT No. C1

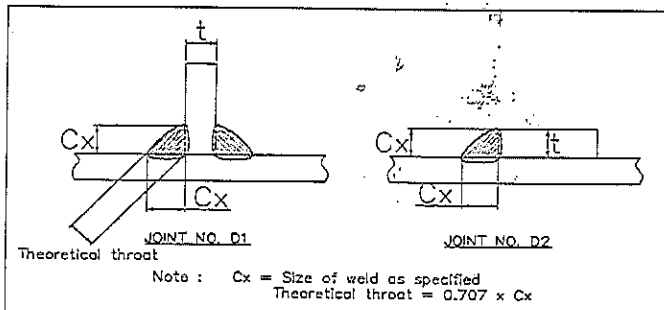


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

JOINT No. B5



Procedure Qualification Record (PQR)

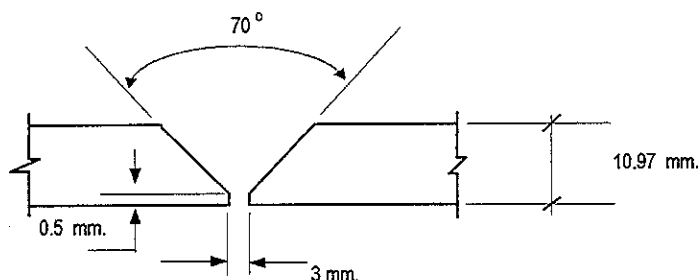
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : KWC-06A0017-PQR-RNC-PN1

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	KWC-06A0017-PQR-RNC-PN1	Date :	23 February 2013
WPS Number :	KWC-06A0017-WPS-RNC-PN1		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding (SMAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)											
Material Spec.		A 106	To	A 106		Temperature		N/A							
Type or Grade		Gr.B	To	Gr.B		Hold Time		N/A							
P. No.		1	To	1		Others									
Thickness of Test Coupon			10.97 mm.												
Diameter of Test Coupon			Ø 6"												
Others															
FILLER METALS (QW-404)				GAS (QW-408)											
				Percent Composition											
					Gas (es)		Mixture		Flow Rate						
				Shielding	Argon		99.99%		8 L/Min.						
				Trailing	-		-		-						
				Backing	-		-								
SFA Specification				GTAW		SMAW		ELECTRICAL CHARACTERISTICS (QW-409)							
AWS Classification				A 5.18		A 5.1		Current		DC (GTAW)		DC (SMAW)			
Filler Metal F-No.				6		4		Polarity		EN		EP			
Weld Metal Analysis A-No.				1		1		Amperage		94 - 150 (GTAW)		Voltage		10 - 12 (GTAW)	
Size of Filler Metal				Ø 2.4 mm.		Ø 3.2 mm.				100 - 112 (SMAW)				23 - 24 (SMAW)	
Others				Kobelco TGS-50		Kobelco LB-52		Tungsten Electrode Size		Ø 2.4 mm. 2% thoriated.EWTh-2					
Weld Metal Thickness				-				Others		N/A					
POSITION (QW-405)								TECHNIQUE (QW-410)							
Position of Groove				6G				Travel Speed				4.73 - 15.24 cm/min.			
Progression of Welding (Uphill or Down)				Up Hill				String or Weave Bead				Both			
Others				N/A				Oscillation				N/A			
PREHEAT (QW-406)								Multiple or single Pass				Multiple Pass			
Preheat Temperature				37° C				Multiple or Single Electrodes				Multiple Electrodes			
Interpass Temperature (max.)				188° C				Others				N/A			
Others				N/A											

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- KWC-06A0017-PQR-RNC-PN1

Tensile Test Report No.TS-13-02-084

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
545/13 TR1	19.05	9.78	186.31	97.06	520.97	OUT OF WELD
545/13 TR2	19.02	9.88	187.92	97.62	519.46	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-02-033

Type and Figure Number	Type of Bend	Indication	Results
545/13 BF1	Transverse face bend	No Open Discontinuity	Accepted
545/13 BF2	Transverse face bend	No Open Discontinuity	Accepted
545/13 BR1	Transverse root bend	2.12,1.57,1.78 (WELD ZONE)	Accepted
545/13 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.WQT-009

Hardness Testing Report No. HV-13-02-015

Macrostructure Report No. MA-13-02-049

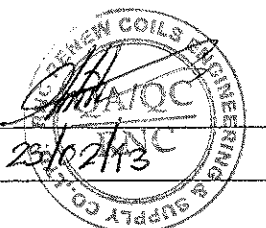
Impact Testing Report No.IM-13-02-092

Welders Name Mr.Prathom M. Welder Number W-009
Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date :



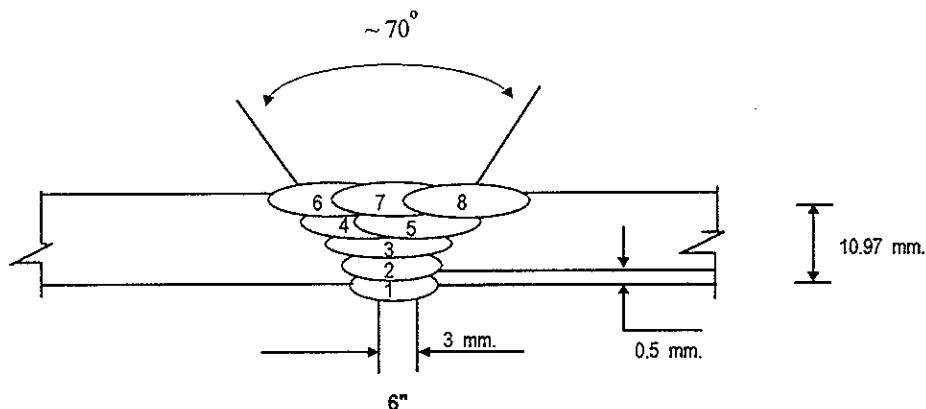
Inspected by : Mr.Wichit K.

Certified by : Mr.Chaloemkiat R.

Date : 22 February 2013

PAE TECHNICAL SERVICE	WELDING PROCEDURE	PQR No. : KWC-06A0017-PQR-RNC-PN1
		WPS No. : KWC-06A0017-WPS-RNC-PN1
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST CONDITION AND RESULT

Welding Sequences for KWC-06A0017-PQR-RNC-PN1 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or Weave	Wire Feed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Bead	Speed				
1	GTAW	ER70S-G	2.4	94	10	DCEN	-	-	4.73	11.93	8	-
2	GTAW	ER70S-G	2.4	150	12	DCEN	-	-	8.88	12.17	8	-
3	SMAW	E7016	3.2	100	23	DCEP	-	-	11.71	11.79	-	-
4	SMAW	E7016	3.2	110	24	DCEP	-	-	15.24	10.40	-	-
5	SMAW	E7016	3.2	111	24	DCEP	-	-	9.51	16.81	-	-
6	SMAW	E7016	3.2	110	24	DCEP	-	-	11.68	13.57	-	-
7	SMAW	E7016	3.2	112	24	DCEP	-	-	9.51	16.96	-	-
8	SMAW	E7016	3.2	111	24	DCEP	-	-	7.39	21.63	-	-

WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1

Client :		Project Name : PQR		Location : RNC Work Shop		Report No. PQR-017																																																																																																																							
Renew Coils Engineering & Supply Co.,Ltd.		Customer Order No. : N/A		WPS No. : KWC-0610017-WPS-RNC-PN1		Date : 13-Feb-13																																																																																																																							
Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results																																																																																																																					
					Spec	Size		Visual	RT																																																																																																																				
W-009	MR.PRATHOM M.	GTAW / SMAW	6G	ASME IX	A106 Gr.B	ø 6" Thk. 10.97 mm.	A5.18 / A5.1	ACCEPTED	ACCEPTED																																																																																																																				
Joint Details <table border="1"> <thead> <tr> <th rowspan="2">Layer (S)</th> <th rowspan="2">Process</th> <th colspan="2">Filler Metal</th> <th colspan="2">Current</th> <th rowspan="2">Travel Speed cm/min</th> <th rowspan="2">Interpas Temp °C</th> <th rowspan="2">Heat Input (KJ / cm)</th> <th colspan="2">Shielding</th> <th rowspan="2">Backing</th> </tr> <tr> <th>Class</th> <th>Dia. (mm.)</th> <th>Type</th> <th>Polarity</th> <th>Amp</th> <th>Volt</th> <th>GAS</th> <th>Flow Rate (l/min)</th> </tr> </thead> <tbody> <tr><td>1</td><td>GTAW</td><td>ER70S-G</td><td>2.4</td><td>DCEN</td><td></td><td>94</td><td>37</td><td>11.93</td><td>Argon</td><td>8</td><td>-</td></tr> <tr><td>2</td><td>GTAW</td><td>ER70S-G</td><td>2.4</td><td>DCEN</td><td></td><td>150</td><td>119</td><td>12.17</td><td>Argon</td><td>8</td><td>-</td></tr> <tr><td>3</td><td>SMAW</td><td>E7016</td><td>3.2</td><td>DCEP</td><td></td><td>100</td><td>123</td><td>11.79</td><td>-</td><td>-</td><td>-</td></tr> <tr><td>4</td><td>SMAW</td><td>E7016</td><td>3.2</td><td>DCEP</td><td></td><td>110</td><td>143</td><td>10.40</td><td>-</td><td>-</td><td>-</td></tr> <tr><td>5</td><td>SMAW</td><td>E7016</td><td>3.2</td><td>DCEP</td><td></td><td>111</td><td>150</td><td>16.81</td><td>-</td><td>-</td><td>-</td></tr> <tr><td>6</td><td>SMAW</td><td>E7016</td><td>3.2</td><td>DCEP</td><td></td><td>110</td><td>153</td><td>13.57</td><td>-</td><td>-</td><td>-</td></tr> <tr><td>7</td><td>SMAW</td><td>E7016</td><td>3.2</td><td>DCEP</td><td></td><td>112</td><td>180</td><td>16.96</td><td>-</td><td>-</td><td>-</td></tr> <tr><td>8</td><td>SMAW</td><td>E7016</td><td>3.2</td><td>DCEP</td><td></td><td>111</td><td>188</td><td>21.63</td><td>-</td><td>-</td><td>-</td></tr> </tbody> </table>										Layer (S)	Process	Filler Metal		Current		Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	Class	Dia. (mm.)	Type	Polarity	Amp	Volt	GAS	Flow Rate (l/min)	1	GTAW	ER70S-G	2.4	DCEN		94	37	11.93	Argon	8	-	2	GTAW	ER70S-G	2.4	DCEN		150	119	12.17	Argon	8	-	3	SMAW	E7016	3.2	DCEP		100	123	11.79	-	-	-	4	SMAW	E7016	3.2	DCEP		110	143	10.40	-	-	-	5	SMAW	E7016	3.2	DCEP		111	150	16.81	-	-	-	6	SMAW	E7016	3.2	DCEP		110	153	13.57	-	-	-	7	SMAW	E7016	3.2	DCEP		112	180	16.96	-	-	-	8	SMAW	E7016	3.2	DCEP		111	188	21.63	-	-	-
Layer (S)	Process	Filler Metal		Current		Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding			Backing																																																																																																																	
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7	SMAW	E7016	3.2	DCEP		112	180	16.96	-	-	-																																																																																																																		
8	SMAW	E7016	3.2	DCEP		111	188	21.63	-	-	-																																																																																																																		

PAE Inspection By SIGNATURE: NAME: Mr. Wichit K. DATE: 14-Feb-2013		Witness By SIGNATURE: NAME: Mr. Chatchawan S. DATE: 19/02/13		Approved By SIGNATURE: NAME: Phuchin Seehomchai DATE:	
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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnrm.co.th

29780

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-02-084

Revision No. 0

Test Product	WPS No. KWC-06A0017-WPS-RNC-PN1	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Test	Reduced Section Tension
Dimension	Ø 6" (10.97 mm Thk.)	Equipment's Capacity	100 TONS
Date Received	19 February 2013	Equipment / Serial No.	INSTRON / 1000DXR1414
Date Tested	20 February 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	545/13 TR1	N/A	9.78	19.05	186.31	N/A	N/A	N/A	97.06	520.97	N/A	OUT OF WELD
2	545/13 TR2	N/A	9.88	19.02	187.92	N/A	N/A	N/A	97.62	519.46	N/A	OUT OF WELD

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

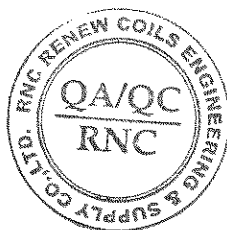
Welder No. : W-009

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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4. This report is tested for LPN Metallurgical Research Center (Thailand) Co.,Ltd.

HQ-F-QA-4-1604

Rev.001/18 n.w. 2556



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmpm.co.th



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Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-02-033

Revision No. 0

Test Product	WPS No. KWC-06A0017-WPS-RNC-PN1	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 6" (10.97 mm Thk.)	Diameter of Meudrel	40 mm
Date Received	19 February 2013	Bend Angle	180 Degree.
Date Tested	20 February 2013	Equipment/ Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	545/13 BF1	Transverse face bend	10.01x38.10x162.51 mm	No Open Discontinuity	N/A	-
2	545/13 BF2	Transverse face bend	10.00x38.05x161.70 mm	No Open Discontinuity	N/A	-
3	545/13 BR1	Transverse root bend	10.25x38.05x163.03 mm	Open Discontinuity	WELD ZONE	2.12, 1.57, 1.78 mm
4	545/13 BR2	Transverse root bend	10.02x37.97x161.18 mm	No Open Discontinuity	N/A	-

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

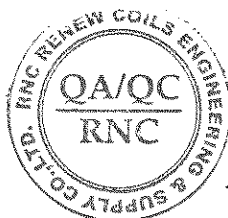
Welder No. : W-009

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

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9792

TESTING
No. 0154

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-02-092

Revision No. 0

Test Product	WPS No. KWC-06A0017-WPS-RNC-PN1	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 and ASTM E23M-07a ^{e1}
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Test	Charpy V-Notch Impact
Dimension	Ø 6" (10.97 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Received Date	19 February 2013	Equipment 's Capacity	542 Joules
Tested Date	20 February 2013	Ambient Temperature	23 ± 5 °C

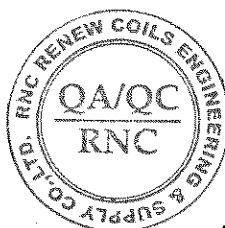
Item No.	Test No.	Test Temperature (°C)	Specimen Dimension			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	545/13-1	-46°C	7.507	10.007	54.81	147.11	199.65	Weld Metal	-
2	545/13-2	-46°C	7.505	10.007	54.79	262.94			
3	545/13-3	-46°C	7.506	10.008	54.86	188.90			
4	545/13-1	-46°C	7.506	10.009	54.92	7.52	9.51	Base Metal	-
5	545/13-2	-46°C	7.506	10.008	54.95	11.32			
6	545/13-3	-46°C	7.506	10.006	54.96	9.70			
7	545/13-1	-46°C	7.507	10.006	54.85	91.68	90.00	HAZ	-
8	545/13-2	-46°C	7.507	10.008	54.88	93.30			
9	545/13-3	-46°C	7.505	10.008	54.78	85.02			

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

Welder No. : W-009



Reported by :

(Mr. Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by :

(Mr. Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

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Macrostructure Analysis Report

Page 1 of 1

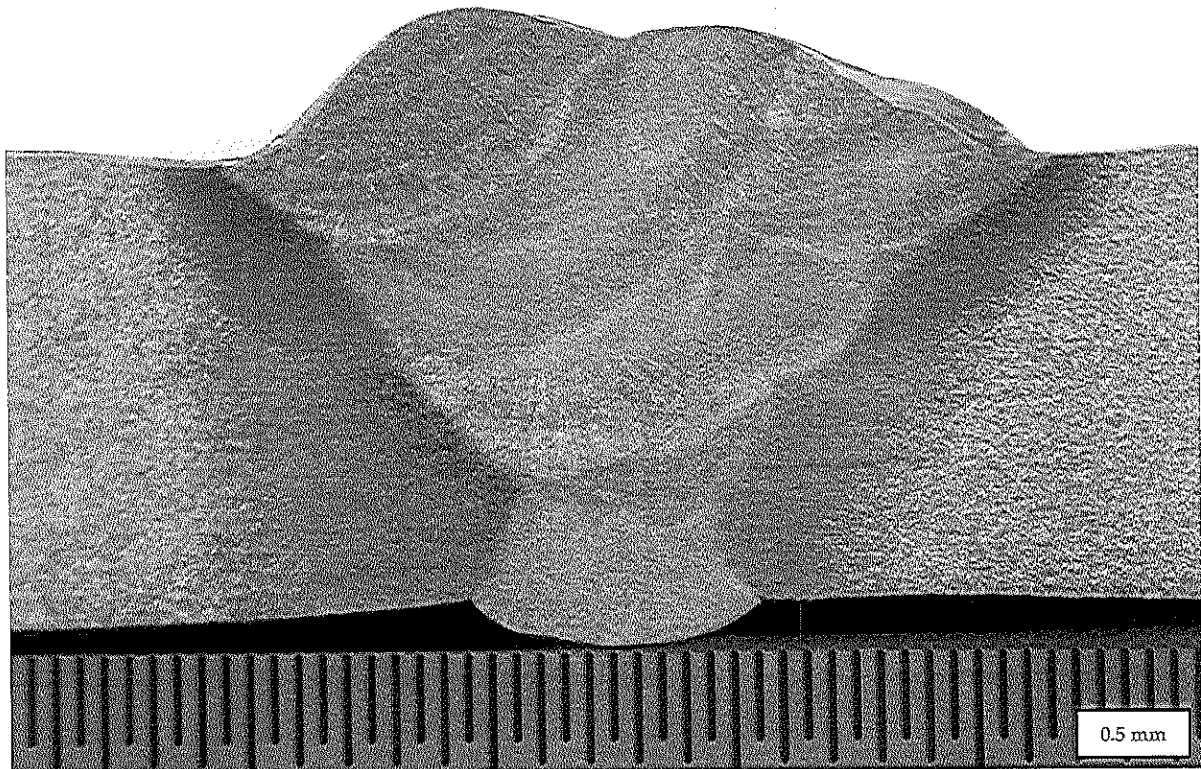
Report No. MA-13-02-049

Revision No.0

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

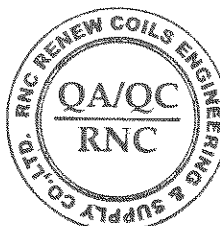
Test Product	WPS No. KWC-06A0017-WPS-RNC-PN1	Material Characterization	Pipe
Test No.	545/13 MA	Equipment used	Microscope upto 40X (times)
Test Location	-	Serial No.	LEICA 0650854912
Material Specification	A106 Gr.B	Method of Preparation	ASME Section IX : 2010 edition
Dimension	6" (10.97 mm Thk.)	Test Method	ASME Section IX : 2010 edition
Welding Process/Position	GTAW+SMAW/6G	Standard of Test	ASME Section IX : 2010 edition
Type of Joint	Butt Jiont	Etchant Reagent	25 ml HNO ₃ + 75 ml H ₂ O
Welder Name / No.	Mr. Prathom M. / W-009	Time of Etching	5 minutes
Date Received	18 February 2013	Date Tested	20 February 2013



Macro-etch Examination : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

Additional details : -

Reported by : 
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 22/02/13



Approved by : 
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 22/02/13

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13232

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

Hardness Testing Report

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Page 1 of 1

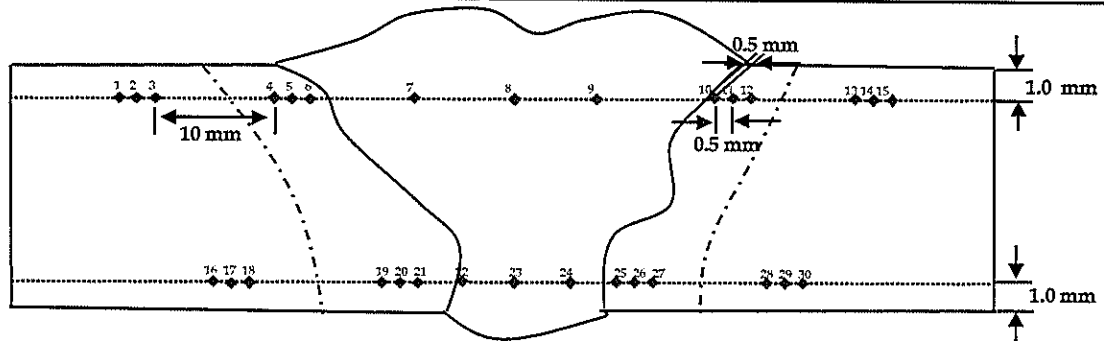
Report No. HV-13-02-015

Revision No. 0

Test Product	WPS No. KWC-06A0017-WPS-RNC-PN1	Material Characterization	Pipe
Test No.	545/13 HN	Test Method	ASTM E92-82 (Reapproved 2003) ^{E2}
Project Name	WELDER SUPPLY AND PQR	Standard of	ASME Section IX : 2010 edition
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A106 Gr.B	Equipment	Wilson Wolpert , 432 SVD (V32D219)
Dimension	6" (10.97 mm Thk.)	Indenter Size	Pyramid diamond, angle of 136°
Welding Process/Position	GTAW+SMAW/6G	Test Load	HV 10 Kgf.
Type of Joint	Butt Jiont	Total Force Dwell Time	15 s
Date Received	18 February 2013	Cal. Block	No. A60023 No. A62072
Date Tested	20 February 2013	Ambient Temperature	26.0 °C

Test Location

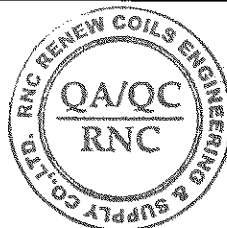
Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale)
1	BASE	142.0	16	BASE	148.5	*****		
2	BASE	141.3	17	BASE	147.5			
3	BASE	140.8	18	BASE	148.3			
4	HAZ	147.8	19	HAZ	180.4			
5	HAZ	160.8	20	HAZ	176.7			
6	HAZ	175.0	21	HAZ	176.2			
7	WELD	180.4	22	WELD	170.0			
8	WELD	166.0	23	WELD	164.7			
9	WELD	164.7	24	WELD	167.0			
10	HAZ	180.8	25	HAZ	175.9			
11	HAZ	164.5	26	HAZ	176.9			
12	HAZ	155.9	27	HAZ	178.6			
13	BASE	139.5	28	BASE	143.9			
14	BASE	139.9	29	BASE	141.9			
15	BASE	138.0	30	BASE	142.7			

Additional details :-

Reported by : 
 (Mr.Keerati Sa-ngoen)
 (Acting) General Manager - Quality Assurance
 Date : 22/02/13



Approved by : 
 (Mr.Pavaret Freedawaphat)
 Deputy Managing Director
 Date : 22/02/13

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